Highest quality genomes we will use (scaffold N50's around 100Mb+, single digit L50's)

Species (assembly level, # scaffolds, scaff N50, scaff L50, year)	**Accession Number**	**Sequencing Method**	**Annotation info**
Ixodes scapularis (chrom, 1k, 208Mb, 5, 2026)	GCA_016920785.2	(Nuss) Pacbio + HiC + Illumina correction	RefSeq NCBI
Ixodes ricinus (chrom, 17k, 168Mb, 6, 2025)	GCA_964199275.3	(de Araujo) 10x chromium illumina	genoscope NCBI
Ixodes pacificus (chrom, 94k, 74Mb, 10, 2025)	GCA_964199305.2	(de Araujo) 10x chromium illumina	genoscope NCBI
Ixodes persulcatus (chrom, 89k, 98Mb, 9, 2025)	GCA_964199295.2	(de Araujo) 10x chromium illumina	genoscope NCBI
Dermacentor reticulatus (chrom, 11, 223Mb, 4, 2026)	GCA_060856085.1	(Kostarnoy) nanopore + pacbio?	NF Gamaleya Federal Research Center NCBI
Dermacentor reticulatus (chrom, 1k, 217Mb, 4, 2026)	GCA_057332255.1	(Billows) single adult nanopore – Hifiasm	none provided but should be able to dig up RNA data
Dermacentor reticulatus (chrom, 22k, 185Mb, 5, 2026)	GCA_059058845.1	(Billows) single adult nanopore – Flye	none provided but should be able to dig up RNA data
Dermacentor variabilis (chrom, 1k, 207Mb, 5, 2025)	GCA_050947875.1	(Luecke) Illumina + PacBio	RefSeq NCBI
Dermacentor andersoni (chrom, 1k, 206Mb, 5, 2024)	GCA_023375885.3	(Tompkin) PacBio + Hi-C	RefSeq NCBI apparently for previous iteration, current on Ensembl with GeneBuild?
Dermacentor silvarum (chrom, 2k, 190Mb, 5, 2022)	GCA_013339745.2	(Jia) Illumina and pacbio + Hi-C	RefSeq and TIGMIC on NCBI, as well as GeneBuild on Ensembl (all current)
Dermacentor albipictus (chrom, 1k, 181Mb, 4, 2024)	GCA_038994185.2 (alternate haplotype GCA_038994205.1)	(Saelao) Illumina + PacBio	RefSeq NCBI
Haemaphysalis longicornis (chrom, 4k, 205Mb, 5, 2020)	GCA_013339765.2	(Jia) Illumina and pacbio + Hi-C	TIGMIC NCBI apparently on previous iteration? current by VEuPathDB on Ensembl?
Haemaphysalis longicornis (scaff, 300, 213Mb, 6, 2025)	GCA_048455015.1	(Moustafa) PacBio + Hi-C	RefSeq NCBI
Haemaphysalis longicornis (chrom, 2k, 235Mb, 13, 2026)	GCA_055474425.1	(Wang) PacBio + Hi-C	Beijing Institute NCBI

- TRIPLOID though			
Amblyomma maculatum (chrom, 200, 189Mb, 5, 2026)	GCA_055883805.1	(Luecke) Illumina + PacBio	RefSeq NCBI
Amblyomma mixtum (chrom, 200, 206Mb, 4, 2026)	GCA_055824585.1	(Saelao) PacBio + Illumina	RefSeq NCBI
Amblyomma americanum (chrom, 400, 212Mb, 5, 2025)	GCA_052857255.1	(Luecke) Illumina + PacBio	RefSeq NCBI
Amblyomma variegatum (chrom, 700, 1Gb, 4, 2026)	GCA_965362215.1	(Liu) unknown?	University of Liverpool UK NCBI
Hyalomma asiaticum (chrom, 6k, 137Mb, 5, 2020)	GCA_013339685.2	(Jia) Illumina and pacbio + Hi-C	TIGMIC NCBI apparently on previous iteration? current by VEuPathDB on Ensembl?
Rhipicephalus microplus (chrom, 1k, 243Mb, 5, 2024)	GCA_043290135.1	(Tidwell) PacBio	RefSeq NCBI
Rhipicephalus sanguineus (chrom, 2k, 208Mb, 5, 2022)	GCA_013339695.2	(Jia) Illumina and pacbio + Hi-C	TIGMIC and RefSeq NCBI

Medium quality genomes we most likely want to use (scaffold N50's in the 2Mb-400kb range)

Species (assembly level, # scaffolds, year)	**Accession Number**	**Sequencing Method**	**Annotation info**
Ixodes hexagonus (chrom, 98k, 1Mb, 64, 2025)	GCA_964199285.2	(de Araujo) 10x chromium illumina	genoscope NCBI
Ixodes persulcatus (scaff, 12k, 532kb, 936, 2020)	GCA_013358835.2	(Jia) Illumina and pacbio (no Hi-C for this one)	TIGMIC NCBI apparently on previous iteration? current by VEuPathDB on Ensembl?
Ixodes inopinatus (contig, 12k, 781kb, 1266, 2024)	GCA_964198085.1	(Baede) Nanopore	none
Ixodes inopinatus (contig, 12k, 756kb, 1305, 2024)	GCA_964198045.1	(Baede) Nanopore	none
Ixodes inopinatus (contig, 14k, 522kb, 1951, 2024)	GCA_964198095.1	(Baede) Nanopore	none
Dermacentor variabilis (contig, 22k, 445kb, 1285, 2025)	GCA_049308735.1	(Cassens) Nanopore of 10 pooled females	none
Haemaphysalis longicornis (scaff, 7k, 1Mb, 820, 2022)	GCA_022414705.1	(Yu) PacBio + Illumina error correction	none
Rhipicephalus microplus (contig, 11k, 2Mb, 316, 2024)	GCA_030522445.2	(Meiring) Nanopore after phenol chloroform extract	Clinglobal NCBI
Rhipicephalus appendiculatus (contig, 33k, 680kb, 607, 2024)	GCA_030522465.2	(Meiring) Nanopore after phenol chloroform extract	Clinglobal NCBI
Rhipicephalus annulatus (contig, 16k, 437kb, 1685, 2020)	GCA_013436015.1	(Guerrero) PacBio	none

Lower quality genomes we might still use (scaffold N50's around 100kb and below)

Species (assembly level, # scaffolds, year)	**Accession Number**	**Sequencing Method**	**Annotation info**
Ixodes persulcatus (scaff, 94k, 84kb, 4379, 2025)	GCA_965286795.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Dermacentor niveus (contig, 127k, 85kb, 4538, 2025)	GCA_965286785.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Dermacentor silvarum (contig, 155k, 54kb, 7427, 2025)	GCA_965286685.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Haemaphysalis concinna (contig, 248k, 26kb, 16874, 2025)	GCA_965286775.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Haemaphysalis longicornis (contig, 298k, 33kb, 16549, 2025)	GCA_965286675.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Haemaphysalis montgomeryi (contig, 346k, 6kb, 63245, 2025)	GCA_965286705.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Haemaphysalis qinghaiensis (contig, 516k, 5kb, 43162, 2025)	GCA_965286765.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Haemaphysalis bispinosa (scaff, 889k, 4kb, 140560, 2025)	GCA_052819225.1	(Amin) Illumina?	none
Amblyomma americanum (scaff, 37k, 116kb, 6011, 2024)	GCA_030143305.2	(Chou) 50 adult females PacBio	Arcadia Science, available on NCBI and Ensembl
Amblyomma geoemydae (contig, 176k, 38kb, 12753, 2025)	GCA_965286655.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Amblyomma javanense (contig, 232k, 55kb, 11509, 2025)	GCA_965286815.2	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Amblyomma testudinarium	GCA_965286825.1	(Du et al. 2025) Nano	none

(contig, 428k, 49kb, 13165, 2025)		+ Ill polish + integrate de novo Ill assembly	
Amblyomma varanense (contig, 416k, 15kb, 23560, 2025)	GCA_965286715.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Hyalomma anatolicum (contig, 130k, 152kb, 2521, 2025)	GCA_965286835.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Hyalomma asiaticum (contig, 129k, 72kb, 4399, 2025)	GCA_965286805.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Hyalomma scupense (contig, 117k, 61kb, 5843, 2025)	GCA_965286695.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Rhipicephalus microplus (contig, 158k, 60kb, 8743, 2025)	GCA_965286725.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Rhipicephalus pumilio (contig, 207k, 50kb, 9097, 2025)	GCA_965286645.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Rhipicephalus sanguineus (contig, 114k, 143kb, 4291, 2025)	GCA_965286755.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none
Rhipicephalus turanicus (contig, 176k, 75kb, 6417, 2025)	GCA_965286665.1	(Du et al. 2025) Nano + Ill polish + integrate de novo Ill assembly	none

Genome groups and their publications:

- (Nuss et al. 2023) The highly improved genome of I. scapularis
    - Big PacBio of females and males -> hifiasm to generate big assembly and Khaper algorithm to reduce redundancy -> scaffold with CHiCAGO and Hi-C of egg masses -> iterative refinement with Pilon with the Illumina reads?
    - https://doi.org/10.26508/lsa.202302109
    - https://www.nature.com/articles/s41588-021-00895-y and https://doi.org/10.5281/zenodo.4780792 for Khaper algorithm, https://doi.org/10.1111/1755-0998.13020 for improving Illumina assemblies with Hi-C and long reads
- (de Araujo et al. 2024) Genome sequences of four Ixodes species expands understanding of tick evolution
    - Ixodes hexagonus ricinus persulcatus pacificus
    - 10X amplicons paired end Illumina (Chromium linked reads?) -> supernova assembler -> Hi-C scaffolding using I. ricinus chromosome assembler
    - https://doi.org/10.1186/s12915-025-02121-1
- (Jia et al. 2020) Large scale comparative analyses of tick genomes
    - Ixodes persulcatus, Haemaphysalis longicornis, Dermacentor silvarum, Hyalomma asiaticum, Rhipicephalus sanguineus, Rhipicephalus microplus
    - Pooled 50-100 larvae from a single female -> Illumina and pacbio + HiC (all except I. persulcatus)
    - https://doi.org/10.1016/j.cell.2020.07.023
- (Baede et al. 2024) Similarities between I. ricinus and I. inopinatus genomes
    - I'm just using their I. inopinatus
    - Individual ticks nanopore'd -> de novo assembly
    - https://doi.org/10.1016/j.crpvbd.2024.100229
- (Kostarnoy 2026 not yet published) D. reticulatus whole genome shotgun project
    - Nanopore and PacBio apparently assembled with hifiasm
    - https://www.ncbi.nlm.nih.gov/nuccore/3387724418
- (Billows et al. 2026) First genome of D. reticulatus
    - Single tick Nanopore, with RNA extraction for their annotation
    - Steps here for a Nanopore assembly
    - Hifiasm assembly (GCA_057332255.1) and then also Flye assembly (GCA_059058845.1)?
    - The paper specifically refers to GCA_057332255.1
    - https://doi.org/10.1186/s12915-026-02646-z
- (Luecke et al. 2025) Genetics of Veterinary Pests - Whole Genome Assembly of the American dog tick, Dermacentor variabilis
    - This is a dataset publication, no paper
    - Illumina and PacBio 18x genome coverage?
    - https://www.ars.usda.gov/research/publications/publication/?seqNo115=427355
    - https://www.ncbi.nlm.nih.gov/nuccore/JBNOKL000000000
- (Cassens et al. 2025) The genome of D. variabilis
    - Nanopore of a pool of 10 ticks
    - This article has assembly steps and Orthofinder usage
    - https://doi.org/10.1093/g3journal/jkaf130
- (Tompkin et al. 2026 preprint) Dermacentor andersoni genome sequencing
    - Read this paper more thoroughly
    - PacBio with Hi-C of single colony adult, 43x coverage

    - https://www.biorxiv.org/content/10.64898/2026.04.15.717773v1.full
- (Saelao et al. 2024) Genetics of Veterinary Pests – Dermacentor albipictus genome version 2
    - This is a dataset publication, no paper
    - Single unfed adult from Rhodes 1998 colony, PacBio and Illumina, 31x coverage
    - https://www.ars.usda.gov/research/publications/publication/?seqNo115=420534
    - https://www.ncbi.nlm.nih.gov/nuccore?LinkName=bioproject_nuccore_wgsmaster&from_uid=1072088
- (Moustafa et al. 2025) Genome of North American Haemaphysalis longicornis
    - This was with the USDA as well and has info on previous assemblies for H. longicornis
    - PacBio + Hi-C
    - https://doi.org/10.1186/s12864-025-11477-1
- (Wang et al. 2025) Triploidy in Haemaphysalis longicornis
    - This is a TRIPLOID genome, unsure how we will be using it
    - PacBio and Hi-C with multiple individuals from a single mother
    - https://doi.org/10.1038/s42003-025-09153-x
- (Chou et al. 2023) De novo assembly of a long-read Amblyomma americanum tick genome
    - PacBio of 25 or 50 unfed ticks, maybe their salivary glands?
    - https://thestacks.org/publications/dataset-amblyomma-americanum-genome
    - https://www.ncbi.nlm.nih.gov/nuccore/2725487107
- (Luecke et al. 2025) Exploiting Veterinary Pest Genomics to Improve Animal Health – Whole genome assembly of the Lone Star Tick, Amblyomma americanum in the National Center for Biotechnology Information
    - This is a dataset publication, no paper
    - Single adult male, PacBio and Illumina with Hifiasm assembly 28x coverage
    - https://www.ncbi.nlm.nih.gov/nuccore/JBQHYB000000000.1
    - https://www.ars.usda.gov/research/publications/publication/?seqNo115=429778
- (Luecke et al. 2025) Exploiting Veterinary Pest Genomics to Improve Animal Health – Whole genome sequencing and assembly of the Gulf Coast tick, Amblyomma maculatum in the National Center for Biotechnology Information
    - This is a dataset publication, no paper
    - Single adult male, PacBio and Illumina with Hifiasm 16x coverage
    - https://www.ncbi.nlm.nih.gov/nuccore/JBRFLQ000000000.1
    - https://www.ars.usda.gov/research/publications/publication/?seqNo115=433109
- (Saelao et al. 2025) Exploiting Veterinary Pest Genomics to Improve Animal Health – Whole genome sequencing and assembly of the Cayenne tick, Amblyomma mixtum in the National Center for Biotechnology Information
    - This is a dataset publication, no paper
    - Pooled male and female, PacBio and Illumina with Hifiasm 20x coverage
    - https://www.ncbi.nlm.nih.gov/nuccore/JBVVRB000000000.1
    - https://www.ars.usda.gov/research/publications/publication/?seqNo115=433108
- (Liu et al. 2026? 2025? UNKNOWN) Amblyomma variegatum placeholder
    - No information given
    - https://www.ncbi.nlm.nih.gov/nuccore/CBDTPI000000000.1
- (Tidwell et al. 2024) Identifying the sex chromosome and sex determination genes in the cattle tick, Rhipicephalus (Boophilus) microplus
    - Single male PacBio
    - https://doi.org/10.1093/g3journal/jkae234
- (Meiring et al. 2025) Tick genomics through a Nanopore: A low-cost approach for tick genomics

    - Phenol chloroform extraction of 1.25g total eggs from 20 females, then Nanopore sequencing
    - https://doi.org/10.1186/s12864-025-11733-4
- (Guerrero et al. 2021) Raw pacific biosciences and illumina sequencing reads and assembled genome data for the cattle ticks Rhipicephalus microplus and Rhipicephalus annulatus
    - PacBio'ed 1.25g of eggs from 9 single paired matings, assembled with Canu and polished with Arrow then used Purge Haplotigs
    - https://doi.org/10.1016/j.dib.2021.106852

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- (Du et al. 2025) 19 tick species from China (~100k+ contigs each, very broken up) – this is primarily a microbiome study btw, and these are supposedly DRAFT ASSEMBLIES
    - Ixodes persulcatus, Dermacentor niveus + silvarum, Haemaphysalis concinna + longicornis + montgomeryi + qinghaiensis, Amblyomma geomydae + javanense + testudinarium + varanense, Hyalomma anatolicum + asiaticum + scupense, Rhipicephalus microplus + pumilio + sanguineus + turanicus
    - Also Alveonasus lahorensis but this is a soft tick
    - POOLED 15-30 ticks -> de novo nanopore -> illumina polishing -> merge with de novo illumina
    - https://www.nature.com/articles/s41564-025-02119-z

- (Yu et al. 2022) The new Haemaphysalis longicornis genome provides insights into its requisite biological traits
    - PacBio and Illumina error correction of 200 females
    - https://doi.org/10.1016/j.ygeno.2022.110317
- (Amin and Sulabh ?) Not published?
    - Illumina of single adult?
    - https://www.ncbi.nlm.nih.gov/nuccore/JBQRAD000000000.1

Note: look for a paper coming out involving D. reticulatus assemblies GCA_051549955.2, GCA_051549975.2, and GCA_051550115.2 – these are by Dillon, published Sep 2026, scaffold level, ~6-8k scaffolds, ~700kb-1.4Mb scaffold N50 and 274-479 scaffold L50 – Nanopore'd and used Flye to assemble